

Data, Database, DBMS

Data

- Raw facts and figures without context.
- Examples: numbers, names, dates, measurements.
- By itself, data doesn't have meaning until it's processed.

DBMS (Database Management System)

- Software that manages databases.
- Provides tools to store, retrieve, update, and secure data.
- Ensures consistency, integrity, and controlled access.
- Examples: MySQL, Oracle, Microsoft SQL Server, PostgreSQL.

Database

- An organized collection of related data.
- Stores data in a structured way so it can be easily accessed, managed, and updated.
- Examples:
 - A library catalog storing book titles, authors, and ISBNs.
 - A company's customer records with names, addresses, and purchase history.

How Data is organised in a table

Attribute or Column

Name	Regd No	Grade
xyz	123	A
pqr	335	B

Tuple or
Row

Data Types - Numeric

INT – Whole numbers

- *Example:* `age INT` → stores values like `25, 42`

BIGINT – Very large integers

- *Example:* `population BIGINT` → stores values like `1380004385`

DECIMAL(M, D) – Exact fixed-point numbers (great for money)

- *Example:* `salary DECIMAL(10,2)` → stores values like `55000.75`

FLOAT / DOUBLE – Approximate floating-point numbers

- *Example:* `temperature FLOAT` → stores values like `36.6`

BIT(n) – Binary values

- *Example:* `is_active BIT(1)` → stores `0` or `1`

Data Types - Strings

Examples

CHAR(n) – Fixed-length string

- *Example:* `gender CHAR(1)` → stores values like 'M', 'F'

VARCHAR(n) – Variable-length string

- *Example:* `name VARCHAR(50)` → stores values like 'Alice', 'Rahul'

TEXT – Large text data

- *Example:* `description TEXT` → stores product details or blog posts

ENUM – Predefined set of values

- *Example:* `size ENUM('S', 'M', 'L', 'XL')` → stores one of the listed options

SET – Multiple predefined values

- *Example:* `hobbies SET('Reading', 'Music', 'Sports')` → stores 'Reading, Music'

Data Types - Date and Time

Examples

DATE – Stores only the date (YYYY-MM-DD)

- *Example:* `birth_date DATE → 1998-07-15`

DATETIME – Stores date and time (YYYY-MM-DD HH:MM:SS)

- *Example:* `order_time DATETIME → 2026-02-03 08:10:00`

TIMESTAMP – Similar to DATETIME but auto-updates with current time

- *Example:* `last_login TIMESTAMP → 2026-02-03 08:10:00`

TIME – Stores only time (HH:MM:SS)

- *Example:* `meeting_time TIME → 14:30:00`

YEAR – Stores year values (YYYY)

- *Example:* `graduation_year YEAR → 2025`

Data Types - Spatial

Examples

GEOMETRY – Base type for all spatial values

- *Example:* `shape GEOMETRY` → stores any geometric object

POINT – Represents a single location (x, y coordinates)

- *Example:* `location POINT` → (23.45, 87.65)

LINESTRING – Sequence of points forming a line

- *Example:* `road LINESTRING` → (0 0, 10 10, 20 25)

POLYGON – Area defined by multiple points forming boundaries

- *Example:* `park POLYGON` → (0 0, 10 0, 10 10, 0 10, 0 0)

MULTIPOINT / MULTILINESTRING / MULTIPOLYGON – Collections of points, lines, or polygons

- *Example:* `district MULTIPOLYGON` → multiple polygons for different zones

Data Types - JSON

JSON – Stores semi-structured data in JSON format

Example:

```
sql
user_data JSON
```


Create a Table

-- Create a new database

```
CREATE DATABASE school;
```

-- Switch to the database

```
USE school;
```

-- Create a table named 'students'

```
CREATE TABLE students (
```

```
  id INT,
```

```
  name VARCHAR(50),
```

```
  age INT,
```

```
  admission_date DATE,
```

```
  grade ENUM('A','B','C','D','F')
```

```
);
```

-- Unique student ID

-- Student name

-- Student age

-- Date of admission

-- Grade with fixed options

View the Contents of the Table

```
SELECT * FROM students;
```

Insert Values into the Table

-- Insert one record

```
INSERT INTO students (id,name, age, admission_date, grade)
VALUES (1,'Alice', 20, '2026-02-03', 'A');
```

-- Insert multiple records at once

```
INSERT INTO students (name, age, admission_date, grade)
VALUES
('Rahul', 22, '2026-01-15', 'B'),
('Meera', 19, '2026-02-01', 'C'),
('Sanjay', 21, '2026-02-02', 'A');
```